UNIT 1 PRE-TEST

ANSWER KEY

The Real Number System

Answer each question, then check your work.

Name _____ Date _____ Period _____

SCORE 3.0 Basic Understanding

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

1. Write as a decimal: $\frac{3}{8}$

= **0.375**

e.g. 0.5

Solution:

3 ÷ 8 = 0.375. It terminates, so it is rational.

2. Write as a fraction: 0.25

= **1/4**

simplest form, e.g. 1/4

Solution:

0.25 = $\frac{25}{100}$ = $\frac{1}{4}$ in simplest form.

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

3. Classify: √144

Classification: **R**

type R or IR

Solution:

√144 = 12, a whole number, so it is rational.

SCORE 4.0 Target Content

1.1 Converting Rational Numbers

Convert each number as directed. Write fractions in simplest form.

4. Write as a fraction: $0.\overline{6}$ (the 6 repeats)

= **2/3**

simplest form

Solution:

One digit repeats → put it over 9: $\frac{6}{9} = \frac{2}{3}$.

5. Write as a decimal: $2\frac{5}{6}$ (2 and $\frac{5}{6}$)

= **2.833...**

use ... for the repeating part

Solution:

$2\frac{5}{6} = 2 + \frac{5}{6} = 2.8333...$ (the 3 repeats).

1.2 Real Number Classification

Classify the number as Rational (R) or Irrational (IR).

6. Classify: $\sqrt{20}$

Classification: **IR**

type R or IR

Solution:

20 is not a perfect square, so $\sqrt{20}$ is irrational. $\sqrt{20} \approx 4.47...$ never ends and never repeats.

Simplify each expression first, then classify the result.

7. Classify: $\sqrt{3} \times \sqrt{27}$

Classification: **R**

simplify, then R or IR

Solution:

$\sqrt{3} \times \sqrt{27} = \sqrt{81} = 9$, a whole number, so rational.

8. Classify: √32 ÷ √2

Classification: **R**

simplify, then R or IR

Solution:

√32 ÷ √2 = √16 = 4, a whole number, so rational.

1.3 Estimating Non-Perfect Square Roots

Estimate each value to the *nearest tenth* by clicking the number line.

9. Estimate: √40



Solution:

$6^2 = 36$ and $7^2 = 49$, so √40 ≈ 6.32, which rounds to 6.3.

10. Estimate: √95



Solution:

$9^2 = 81$ and $10^2 = 100$, so √95 ≈ 9.747..., which rounds to 9.7.

SCORE *5.0* Advanced Content

1.2 Real Number Classification

Evaluate or simplify each expression if possible, then classify as Rational (R) or Irrational (IR).

11. Classify: −(√49 − √36)

Classification: **R**

simplify, then R or IR

Solution:

−(√49 − √36) = −(7 − 6) = −1, an integer, so rational.

12. Classify: 3 + √5

Classification: **IR**

type R or IR

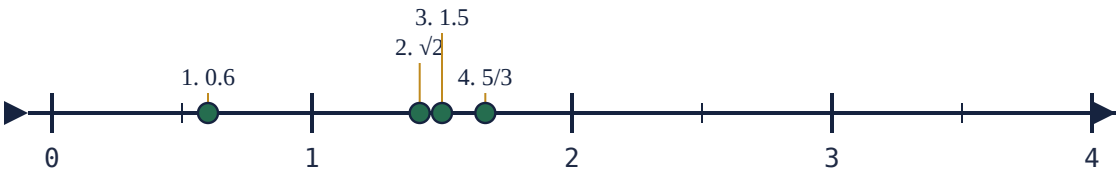
Solution:

√5 is irrational, and a rational number (3) plus an irrational number is irrational.

1.4 Ordering Real Numbers

Click the values in order from *least to greatest*. Each one drops onto the number line.

13. Set: { 0.6, √2, 1.5, $\frac{5}{3}$ }



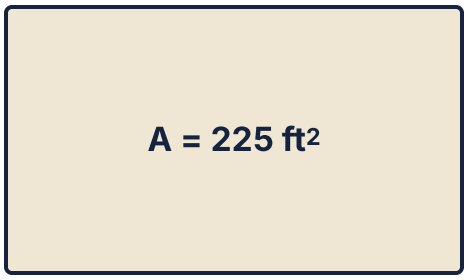
Order: **0.6, √2, 1.5, 5/3**

Solution:

As decimals: 0.6, √2 ≈ 1.41, 1.5, $\frac{5}{3}$ ≈ 1.67 → least to greatest: 0.6, √2, 1.5, $\frac{5}{3}$.

1.5 Area

A square garden has an area of 225 ft². Find the side length and the perimeter.



14. Find the side length and perimeter of the square garden.

Side length = **15 ft**

Perimeter = **60 ft**

Solution:

Side = √225 = 15 ft. Perimeter = 4 × 15 = 60 ft.